APPLICATION OF DINSAR TECHNIQUE TO HIGH COHERENCE SATELLITE IMAGES FOR STRATEGIC INFRASTRUCTURE MONITORING

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ABSTRACT

In this paper the authors present and validate a procedure for bridge monitoring, based on freely available satellite data and the straightforward Differential SAR Interferometry (DInSAR) technique. A displacement dataset of the Morandi bridge in Genoa (Italy) has been created, before its collapse. The outputs obtained were then compared to those found in literature and achieved through the Persistent Scatterer Interferometry (PSI), a more complex and reliable technique. Results demonstrate that the adopted procedure has great potentiality in the application field and could effectively be extended to different types of civil infrastructures.

Index Terms— Bridge monitoring, DInSAR, Copernicus Open-Access, Sentintel-1, ASFVertex

1. INTRODUCTION

As well known, during their life civil structures (bridges, dams or other strategic infrastructures) could modify their behaviour, mainly due to construction material modification, changes in loads, aggressive environmental conditions, design and construction errors [1]. Their increased aging makes urgent quantifying their current and future performance by means of advanced structural health monitoring and damage detection tools [2], [3], to avoid the consequences of an eventual collapse, with the related economic and human losses. Several and different on-site monitoring methods may be adopted according to the specific situation [4]. The contact methods, using accelerometers and fiber-optic, for instance, could be employed to monitor the most critical points of the construction in a very short time. Yet, they cannot be used to monitor the whole bridge because they are time-consuming and expensive [4]. On the other hands, methods based on a limited set of sensors mounted on the structure have the disadvantage of providing incomplete (in space and time) displacement information. Given the above considerations, even though straightforward satellite image processing techniques do not allow distinguishing between stress accumulation or material degradation processes, they could be fundamental in detecting important structural deficiencies and anomalous behaviours. Moreover, the combined use of remote sensing imagery and on-site data could strengthen the quality of the overall monitoring process. In this study, a multidisciplinary approach was adopted by joining the different expertise in civil and telecommunication engineering. The outputs obtained through the DInSAR technique were discussed and compared to the results available in literature on the same case-history and provided by [5], where the more complex PSI technique has been used [6]. Interesting conclusions are given.

2. THE MORANDI BRIDGE

The Morandi Bridge, also called Polcevera viaduct from the underneath river, was designed by the engineer Morandi and built in the 1960s [7], [8], [9]. It was one of the Italian most important bridges, with a prestressed concrete cable-stayed typology [10], connecting the motorways Milan-Genoa and Genoa-Savona, rising above civil and industrial buildings. It was composed of 11 spans (total length of about 1102 meters), supported by two single piers (1 to 2), six V-shaped piers (3 to 8) consisting of two reinforced concrete inclined piers, and A-shaped frames consisting of three independent balanced systems (see Figure 1). The three-span continuous deck were supported by stay cables, made of steel and concrete shells casted around the cables to

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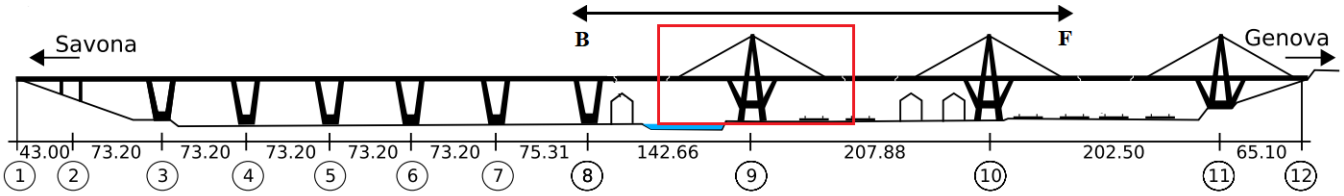


Fig. 1: The bridge longitudinal scheme (in meters) with the collapsed balanced system in the red box.

protect them and to reduce their elongation [9]. On August 14, 2018, one of the three balanced system (9 in Figure 1) collapsed, causing the death of 43 people. As reported in [8] and [10], in the 1990s "during maintenance and repair activities, it was discovered that the stays of the three balanced systems were suffering from widespread general deterioration, as well as several instances of concentrated degradation". In addition, the structure was exposed to marine winds canalized in the valley and struck directly by acid fumes from closeby chimneys. The progressive reduction of both the steel section and the compressed concrete section in the stays could have led to local displacements and deck level irregularities [8].

3. APPLIED PROCEDURE AND RESULTS

In this paper, the authors applied to the Morandi bridge the procedure proposed and already validated in [11], where on-site measurements acquired on strategic infrastructures (dams) were compared to the surface displacements derived from free Sentinel-1 images. In this work, a time-series of interferograms was built, over the window from August 2, 2017 to August 9, 2018, by using data from different sources, like the Copernicus Open-Access Hub by ESA [12] and the ASF Vertex by NASA [13]. It should be pointed out that critical structure monitoring through remote sensing techniques (DInSAR or PSI) allows anomalous displacements to be immediately read remotely, by covering with a high resolution a large area of interest, but the same techniques can be stealthy, since some pixels on the ground can interfere with the pixels on the bridge, resulting in meaningless information. To avoid this impasse, huge amount of data needs to be acquired and analyzed. Luckily, the automated tool developed by [14] comes to help. Details on the adopted procedure are provided hereinafter.

3.1. Data collection and processing

It is worth to emphasizing that the selection of some parameters is needed and some trials necessary to do that. In fact, among all the available images for the same area of interest there are usually multiple acquisitions of the same satellite, but with different trajectories, angles of view and orientation. Data with the same acquisition conditions must be selected, but a careful evaluation of the same is also essential. In the analysis carried out over the one-year time interval, the images acquired during the descending pass were chosen, after verifying that those from the the ascending pass showed very low quality in terms of coherence, result robustness and reliability. For interested readers, the adopted combination of parameters in the present study was the following: 168 for the path; 144 for the frame; and descending for the pass, as explained. Once the images have been selected, a time-series of intergerograms was created through the couples of images, and according to the procedure described in [11], whose interested readers are referred to. Clearly, in this case the master of each couple of images, was the slave of the previous couple, and so on.

3.2. Data cleaning

The final outcome of the proposed processing would be a collection of displacement maps, covering the whole area of interest. However, it is not convenient to consider all the map since the bridge represents a very restricted portion of that region. Consequently, a single line from point A to point G, covering the part of the bridge where the collapse occurred, has been considered to build the time-series displacement map (Figure 2). For this purpose, the QGIS software was adopted to select a well-defined area and then export the geometric features to SNAP (the free ESA tool available for processing the images). SNAP has made it possible read the coordinates of every pixel filtered by the QGIS mask and, after a double-check on Google Maps, it has been discovered

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that not all the pixels were located on the surface of the bridge. For this reason, only 7 points, actually placed on the structure of the bridge, were extracted from the whole set of points, as shown in Figure 2, and moreover, those furthest from the collapsed balanced system (point A and G) were not considered. The rationale behind the

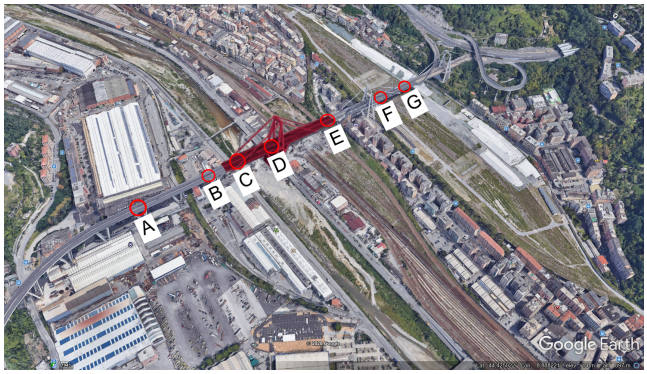


Fig. 2: Selected points on the bridge

choice of the five points (from B to F in Figure 2) is to identify anomalous differential displacements between the two parts of the bridge where the collapse occurred. In other words, as the collapse location is well-known (ex-post), it is worth investigating the displacement evolution and rates in the sections close to the collapsed one.

3.3. Data Analysis

The resulting data set is composed of five time-series of displacements, one for each point selected on the bridge structure, and the response of these points has been interpreted through data visualization. Since all displacements provided by the DInSAR technique are relative to the first master image acquired on August 2, 2017, it is possible to extract the evolution of each pixel, by accumulating its displacements over time. However, since the procedure proposed in this study aims to detect the lowest level of damage, for alerting that some phenomena are happening, only the trend of the displacements is of interest. In particular, observing their variation by starting from August 2017, a cyclic trend until the end of February 2018 is evident, but in March 2018 the variation of the displacements become twice, with a crescent trend. In Figure 3, all the absolute values, s', of the displacements for the five monitored points have been plotted against the local maximum value attained before the end of February 2018. Therefore all the numerical values are dimensionless, and allow to draw the trend of the displacements, for the five points for each date, as represented by the medium dashed line in Figure 3. The change of trend underlines

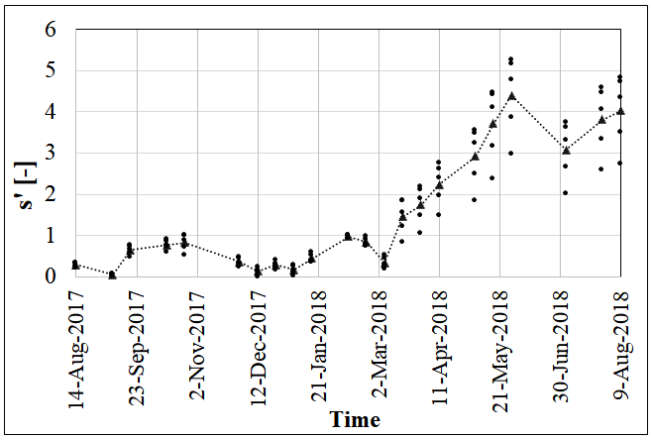


Fig. 3: Dimensionless displacements of the points B-F during the observed period

that a phenomenon was occurring, by confirming the possible use of this simplified technique as a warning of a structure degradation. In Figure 4, the same dimensionless values, s', of each point reached in August 2018 (few days before collapse), are plotted by showing that displacements, up to 5 times the maximum value of the regular trend exhibited in the first monitored period, were reached. The results are in agreement with those provided by other researchers [5], when the more complex PSI technique is used [6]. As general comment

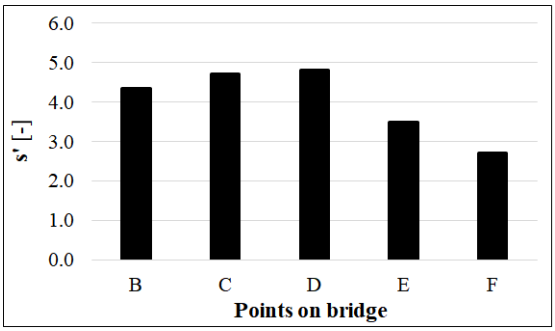


Fig. 4: Final displacements along the bridge on 9 August 2018

of the results in terms of the structure behaviour, the cyclic trend of the first period between August 2017 and February 2018 could be coherent with the seasonal effect of the temperature. Conversely the clear continuous increment of displacements after this period highlights

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degradation phenomena [15], where some damaging effects are likely to be in progress. They might be many: from loss of the prestressing action, to the variation of the static behaviour due to changing in restrains, to the mechanical effect for dynamic repeated action of vehicles' transit, and to geotechnical phenomena at the foundations. It is not possible to establish the causes of the displacement variations without a numerical model of the bridge and a detailed analysis, but it is evident that an unusual structural behaviour has been detected by the applied technique, absolutely useful for creating a warning in a monitoring process.

4. CONCLUSIONS AND FUTURE WORKS

In this paper freely available Sentinel-1 satellite data have been processed to identify the rate of displacements in the parts of the Morandi bridge where collapse occurred. Results demonstrate that the adopted procedure, based on DInSAR technique, has great potentiality in the application field, as it represents a simple and inexpensive method to monitor large structures periodically. The warning of degradation phenomena can be acquired, but further studies are necessary to achieve an higher level of damage detection and comprehension. Future works will employ the SARPROZ software [16] that has been used for some data processing but not extensively in the analysis carried out in this work.

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